Felicity Solar LiFePO4 battery – RS485 register map

Voltronic “LIB” BMS protocol as implemented by Felicity FLB48314TG1 (also FLB48100 / LPBF48 / LPBA48 family). Modbus RTU, 9600 8N1, function 0x03. Verified on three packs in parallel, 2026-10-10. Full description: docs/protocol.md in github.com/KrzysztofHajdamowicz/felicity-battery-modbus.

1. Frames

Direction	Layout	Notes
Request (master → battery)	`ID | 03 | start_hi | start_lo | count_hi | count_lo | CRC_lo | CRC_hi`	Standard Modbus RTU “Read Holding Registers”. CRC-16/Modbus, low byte first.
Response (battery → master)	`ID | 03 | **count_hi | count_lo** | reg0_hi | reg0_lo | … | CRC_lo | CRC_hi`	**Non-standard:** a 16-bit register count replaces the 8-bit byte count. Frame length = 4 + 2·count + 2. Standard Modbus libraries reject these frames; parse them yourself.
Error (document)	`ID | 83 | code | CRC_lo | CRC_hi`	code 1 illegal ID, 2 illegal function, 3 CRC error. Not observed.

Example, address 1, registers 37–53:

```
TX 01 03 00 25 00 11 94 0D
RX 01 03 00 11 00 04 0B C4 0B C4 0B C4 0B BA 00 00 00 00 00 00 00 00 00 00 00 00 00 00 13 D0 A2 00 5F 00 05 57 30 <CRC>
→ 4 sensors, 28.05 / 28.05 / 28.05 / 27.05 °C, charge 0.0 A, discharge 1.9 A, 53.410 V, SOC 95 %, 350.000 Ah
```

2. Register map (FLB48314TG1 scaling)

Reg	Hex	Name	Type	Scale	Unit	Status	Observed / notes
1	0x01	Protocol type	u16	1	–	document	default 0
2	0x02	Protocol version	u16	1	–	document	default 0
3–4	0x03	BMS firmware version	u32	–	–	document	
5–6	0x05	BMS hardware version	u32	–	–	document	
16	0x10	Number of cells	u16	1	–	**verified**	16
17–36	0x11–0x24	Cell voltage 1..20	u16	0.001	V	**verified**	3338–3344 (document says 0.1 V; Felicity uses mV). Cells beyond the count read 0.
37	0x25	Number of temperature sensors	u16	1	–	**verified**	4
38–47	0x26–0x2F	Temperature 1..10	u16	0.1 K	–	**verified**	°C = raw/10 – 273.15. 3012 → 28.05 °C. Sensor 6 = MOSFET (document). Unused sensors read 0.
48	0x30	Charge current	u16	0.1	A	**verified**	0 while discharging
49	0x31	Discharge current	s16	0.1	A	**verified**	0 while charging; use |value|
50	0x32	Pack voltage	u16	0.001	V	**verified**	53400–53470 (document says 0.1 V; Felicity uses mV)
51	0x33	State of charge	u16	1	%	**verified**	95, 96
52–53	0x34	Full-charge capacity	u32	0.001	Ah	**verified**	0x0005 0x5730 = 350000 mAh
54	0x36	Packs online	u16	1	–	document	
55	0x37	Charge alarm	bits	–	–	document	b0 CHT, b1 COV, b2 CLT, b3 OCC
56	0x38	Discharge alarm	bits	–	–	document	b0 DHT, b1 DLT, b2 FETHT, b3 CUV
57	0x39	Charge protect	bits	–	–	document	b0 PRES, b1 BoostNRDY, b2 AFEComm, b3 AFEOCC, b4 AFEOCD, b5 AFESCD, b6 FETHT, b7 SUV, b8 CHT, b9 CLT, b10 OCV, b11 OCC, b12 2NDOVP, b13 CANID, b15 OCC2
58	0x3A	Charge protect 2	bits	–	–	document	b0 ShutdownByCmd, b1 Shutdown, b2 OCCHw, b3 IDError
59	0x3B	Discharge protect	bits	–	–	document	b0 PRES, b1 BoostNRDY, b2 AFEOCC, b3 AFEOCD, b4 AFESCD, b6 IDError, b10 FETHT, b11 CUV, b12 OCD, b13 DLT, b14 DHT, b15 Shutdown

Reg	Hex	Name	Type	Scale	Unit	Status	Observed / notes
60	0x3C	Discharge protect 2	bits	–	–	document	b0 AFEComm, b1 Short, b2 OCD2, b4 2NDOVP, b5 DHT2, b6 ShutdownByCmd, b13 CANID
61	0x3D	BMS state	u16	–	–	document	reserved, 0
62–63	0x3E	Design capacity	u32	0.001	Ah	document	
64–74	0x40–0x4A	Cell count, cell-pair voltage state	u16	–	–	document	00 normal, 01 below limit, 02 above limit
80–85	0x50–0x55	NTC count, sensor-pair temperature state	u16	–	–	document	same coding
96–105	0x60–0x69	Module/cell voltage, current, temperature states	u16	–	–	document	same coding
112	0x70	Charge voltage limit	u16	0.1	V	**verified**	576 → 57.6 V
113	0x71	Discharge voltage limit	u16	0.1	V	**verified**	480 → 48.0 V
114	0x72	Charge current limit	u16	0.1	A	**verified**	320 / 400 / 480 → 32–48 A, differs per pack
115	0x73	Discharge current limit	u16	0.1	A	**verified**	1600 → 160 A
116	0x74	Charge / discharge status	bits	–	–	**verified**	b7 charge enable, b6 discharge enable, b5 charge immediately (SOC ≤ 9 %), b4 charge immediately 2 (SOC 10–14 %), b3 full charge request, b2 small-current charge request. Observed 0xC0.
117	0x75	Run time to empty	u16	1	min	document	
118–119	0x76	Remaining capacity	u32	0.001	Ah	document	

“verified” = observed on FLB48314TG1 and cross-checked against an independent reading of the same packs. “document” = listed in the Voltronic document, not yet confirmed on Felicity firmware (unimplemented registers read 0x0000). All registers are 16-bit big-endian; u32 values are high word first.

3. Derived values

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temperature_n [°C] = raw_n / 10 − 273.15 (raw > 1000; otherwise raw − 273.15)
current [A] = charge_current − |discharge_current| (negative = discharging)
power [W] = current × pack_voltage
pack temperature = mean of sensors 1..3
cell average / min / max / spread over cells 1..number_of_cells with value > 0
```

4. Pack discovery and scaling detection

1. Read register 51 (SOC) from address 0, then 1, 2, … until a pack answers; continue while consecutive addresses answer, stop at the first silent address or at 16. ~500 ms between probes. A zero register count means “blank pack”.
2. Read 16..20: cell 1 > 1000 ⇒ cells in mV. Then read 50: ≥ 10000 ⇒ pack voltage in mV (FLB48314TG1), else 0.1 V. Cell 1 ≤ 1000 ⇒ both in 0.1 V as in the document. Register 50 = 0 ⇒ only the 48..53 block is supported.

5. Suggested polling plan

Block	Registers	Interval per pack	Why
Status	37..53 (17)	2–5 s	temperatures, current, voltage, SOC, capacity
Cells	16..36 (21)	20–30 s	cell voltages change slowly
Limits + status bits	112..116 (5)	1–5 min	limits change with temperature and SOC
Alarms	54..63 (10)	10–30 s	untested on Felicity; read once and look for non-zero

Largest request confirmed to work: 21 registers. Response arrives 60–80 ms after the request; use a 500 ms timeout. About 1 in 10 responses on a USB adapter arrived truncated: validate CRC, drop, re-ask next cycle. One master only on the bus.